

# **REPORT ON AI DEPLOYMENT & AUTOMATION**

Submitted To

**School of Computer Science and Engineering**  
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree  
of  
**B.Tech CSE**



By

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**Batch:** 2024-28

## **CANDIDATE'S DECLARATION**

I, **SAHIL HASHMI** do hereby declare that the internship report titled “**AI Deployment & Automation**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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## **ACKNOWLEDGEMENT**

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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## **INTERNSHIP COMPLETION CERTIFICATE**

The official certificate issued by the organization is appended below:



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# Certificate of Virtual Internship

This is to certify that

sahil hashmi

IILM University, Greater Noida

AI Deployment & Automation Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

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GRADE—O (Outstanding), 90-100 / E (Satisfactory), 80-89 / A (Very Good), 75-79 / B (Good), 65-69 / C (Fair), 60-64 / D (Average), 45-59 / F (Poor), 35-39 / F (Fail), Below 33

## CHAPTER 4: PROJECT DESCRIPTION

### 4.1 Introduction

The rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML) has shifted the industry focus from model development to deployment, operationalization, and automation. The **AI Deployment & Automation Virtual Internship**, supported by Siemens in collaboration with AICTE and EduSkills, focused on bridging the gap between theoretical AI concepts and real-world industrial deployment pipelines. This internship provided comprehensive exposure to integrating automated workflows, managing AI model lifecycles, and scaling intelligent systems efficiently.

### 4.2 Organization Profile

- **Siemens:** A global technology powerhouse focusing on industry, infrastructure, digital transformation, and transport, driving automated solutions worldwide.
- **AICTE (All India Council for Technical Education):** The statutory body and a national-level council for technical education in India under the Ministry of Education.
- **EduSkills:** A non-profit digital-skilling platform that bridges the gap between academia and industry by empowering students with futuristic technical skills.

### 4.3 Problem Statement

Traditional software engineering pipelines often struggle with deploying, monitoring, and maintaining machine learning models in production environments. Issues such as model drift, manual deployment bottlenecks, lack of scalability, and inefficient automation frameworks lead to delayed insights and high operational costs. There is a critical need to implement robust AI deployment structures and automated workflows to streamline production lifecycles.

### 4.4 Project Objectives

- To understand the end-to-end lifecycle of deploying AI and machine learning models into live production environments.

- To explore industrial automation frameworks and strategies supported by Siemens technologies.
- To implement CI/CD (Continuous Integration and Continuous Deployment) concepts tailored for AI workflows.
- To monitor system performance, manage model updates, and minimize manual deployment errors.

#### 4.5 Scope of the Project

The scope of this virtual internship encompasses:

- Studying integration techniques for deploying machine learning pipelines.
- Understanding automated infrastructure management and software deployment frameworks.
- Analyzing real-time tracking, feedback loops, and performance metrics for deployed AI systems.
- Exploring industrial automation standards used in modern software and hardware deployments.

#### 4.6 Technologies and Tools Used

- **AI/ML Frameworks:** Python, Scikit-Learn / TensorFlow fundamentals
- **Automation & Deployment Tools:** CI/CD pipelines, Git version control
- **Virtualization & Environment Management:** Virtual environments, containerization concepts (Docker basics)
- **Collaboration & Learning Platforms:** National Internship Portal, EduSkills LMS

#### 4.7 System Architecture

The deployment architecture follows a modular pipeline structure:

1. **Data Ingestion & Model Training Layer:** Raw operational data is processed, and models are trained and validated.

2. **Packaging & Containerization Layer:** Models and dependencies are wrapped into isolated containers to ensure environment consistency.
3. **Deployment & CI/CD Pipeline:** Automated scripts push the packaged model to staging/production servers upon code validation.
4. **Monitoring & Feedback Loop:** Real-time performance tracking monitors system health, resource utilization, and prediction drift.

#### 4.8 Methodology

- **Phase 1 - Theoretical Foundations:** Covered fundamental architecture patterns for AI integration, workflow automation, and infrastructure deployment strategies.
- **Phase 2 - Hands-on Modules:** Executed guided lab sessions focusing on deployment configurations, script automation, and pipeline management.
- **Phase 3 - Case Studies & Application:** Analyzed industry-grade implementation strategies provided by Siemens to understand large-scale industrial automation.
- **Phase 4 - Evaluation & Assessment:** Completed periodic milestone evaluations to ensure comprehensive understanding of automation lifecycles.

#### 4.9 Expected Outcomes

- Acquired practical proficiency in managing AI deployment workflows and system automation.
- Gained deep insights into industrial digital transformation practices backed by Siemens Industry Software.
- Developed a strong foundation in handling production-level deployment obstacles, automation protocols, and model monitoring frameworks.

#### 4.10 Certificates of Completion and Other Communication Mail Proof

The official virtual internship details and credentials verified upon completion are outlined below:

- **Virtual Internship Program:** AI Deployment & Automation
- **Duration:** April – June 2026
- **Supported By:** Siemens, AICTE (Ministry of Education), and EduSkills



## **CHAPTER 5: BIBLIOGRAPHY / REFERENCES**

1. Siemens Industry Software, *AI Deployment and Automation Curriculum and Training Modules*, EduSkills Virtual Internship Portal, 2026.
2. Artificial Intelligence Council for Technical Education (AICTE), *National Internship Portal Guidelines for Technical Reports*, Ministry of Education, India, 2026.
3. Hands-On Machine Learning Concepts, Deployment Documentation, and CI/CD Pipeline Framework References, EduSkills Learning Management System.